

# E-Waste & Environmental Management

## Activity Worksheet

### The Hidden Carbon Footprint of Our Gadgets

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Date: 05/08/2026

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## Part A: Carbon Footprint Calculation

Use the carbon footprint calculator provided by your faculty.

| Parameter                               | Your Response                       |
|-----------------------------------------|-------------------------------------|
| Annual Carbon Footprint                 | <u>17</u> tCO <sub>2</sub> e / year |
| National Average                        | <u>7</u> tCO <sub>2</sub> e / year  |
| Global Average                          | <u>19</u> tCO <sub>2</sub> e / year |
| Number of Earths Required (if provided) | <u>0.62</u>                         |

find out

Which category contributed the most?

☐ Transportation ☐ Home Energy ☐ Food ☐ Shopping & Consumption

☒ Electronics / Digital Lifestyle ☐ Other: \_\_\_\_\_

Which result surprised you the most?

I was surprised that using electronic devices & replacing them creates more carbon emissions than I expected

## Part B: My Electronic Lifestyle

1. How many smartphones have you owned in the last 10 years?

☐ 1 ☒ 2-3 ☐ 4-5 ☐ More than 5

2. How often do you replace your smartphone?

☐ Every year ☐ Every 2 years ☐ Every 3-4 years ☒ More than 5 years

3. What do you usually do with your old phone?

☒ Keep ☐ Sell ☐ Exchange ☐ Donate

☐ Recycle ☐ Throw away

4. Have you ever repaired an electronic device instead of replacing it?

☒ Yes ☐ No

If yes: Almost 2-3 times charging/replacing mobile display.

5. Have you ever recycled e-waste through an authorized recycler?

☒ Yes ☐ No

Item: Power bank, Charger

| Unused Electronic Items at Home | Number     |
|---------------------------------|------------|
| Mobile phones                   | <u>1</u>   |
| Chargers                        | <u>2</u>   |
| Earphones                       | <u>0</u>   |
| Batteries                       | <u>0</u>   |
| Cables                          | <u>3-4</u> |
| Other                           | <u>-</u>   |

### Part C: Think Like an Environmental Engineer

| Habit                                | Higher Carbon Footprint             | More E-Waste                        |
|--------------------------------------|-------------------------------------|-------------------------------------|
| Buying a new phone every 2 years     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Repairing a laptop                   | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Keeping old phones unused in drawers | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Recycling electronic devices         | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Using devices for 5-6 years          | <input type="checkbox"/>            | <input type="checkbox"/>            |

Which stage of an electronic product's life cycle contributes the MOST carbon emissions?

☐ Raw material extraction ☒ Manufacturing ☐ Transportation

☐ Usage ☐ Disposal

Why is extending the life of an electronic device environmentally beneficial?

It reduces the need to manufacture new devices, saves natural resources, lowers carbon emissions & reduces e-waste.

## Part D: Reflection

Three habits that increase my personal carbon footprint:

1. Charging mobile phones for long hours.
2. Replacing accessories like charger & cables frequently
3. Buying new electronic devices when not necessary

Three habits that increase e-waste generation:

1. Keeping old electronic devices unused.
2. Replacing or buying digital accessories frequently.
3. Throwing damaged devices instead of repairing or recycling them

| Action to Reduce Carbon Emissions & E-waste | Expected Impact                                  |
|---------------------------------------------|--------------------------------------------------|
| Use devices for a longer time               | Reduces carbon emissions & e-waste               |
| Repair devices instead of replacing them    | Save resources and reduces waste                 |
| Recycle old devices/electronics             | Prevents pollution & recovers valuable materials |
|                                             |                                                  |

## Discussion Questions (Optional)

1. Which stage of an electronic product's life cycle has the highest carbon footprint and why?

The manufacturing stage has the highest carbon footprint because it requires raw materials, energy & complex production processes.

2. How does extending the lifespan of smartphones reduce both carbon emissions and e-waste?

Using a smartphone for a longer time reduces the need to manufacture new phones & decreases the amount of electronic waste produced.

3. Why is recycling alone not enough for sustainable e-waste management?

Recycling is important, but, reducing, reusing, & repairing devices first helps save more resource & creates less waste.

4. Explain how the 5Rs contribute to lowering the environmental impact of electronic products.

Recycling is important, but reducing, reusing & Repair & recycle help by reducing waste, conserving natural resources, extending the life of products, & lowering carbon emissions

5. As an engineering student, what innovative solution would you propose to reduce the carbon footprint associated with electronic devices?

I could develop a Smart mobile app that reminds users to repair, reuse, or recycle electronic devices.

The app could also show nearby authorized e-waste collection

centers & reward users with points for responsible disposal

which they can use for shopping, food delivery apps and more e-platforms that they used daily. This would encourage sustainable habits & help reduce carbon emissions & e-waste.

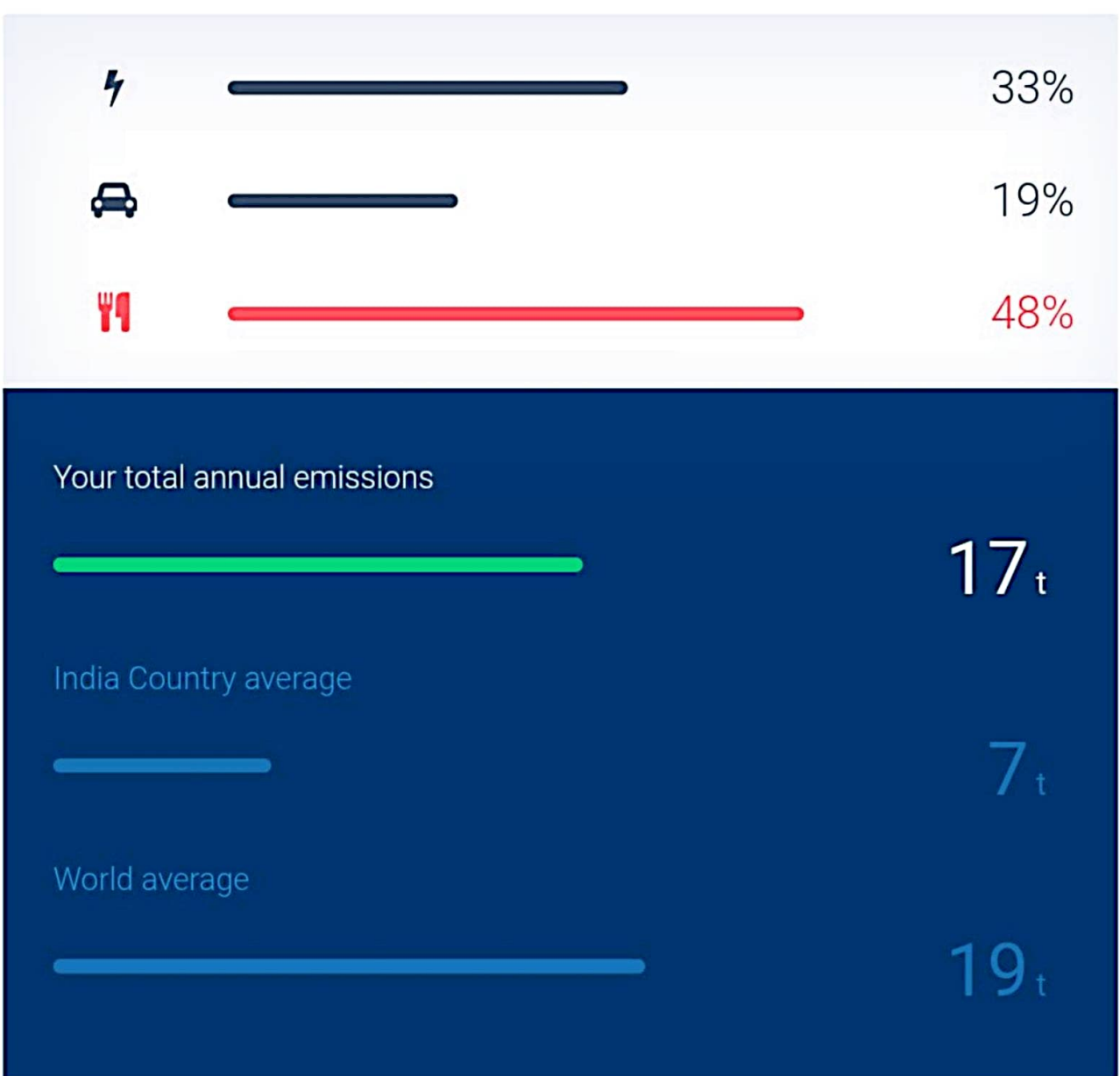
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[← Recalculate](#)

## UN carbon footprint calculator

### Your annual household footprint



Country and world average emissions are shown for comparison. The values are adjusted for the number of household members but do not include international flights. All greenhouse gases are converted into CO<sub>2</sub> equivalent.

Read about our emissions estimation method [here](#).

OFFSET MY FOOTPRINT

### Benefits of offsetting

- Support UN certified green projects that have positive social impact and reduce carbon emissions.
- 100% of the offsetting cost goes directly to the project. Direct contributions have stronger impact.
- Removing emissions permits from the market makes it more expensive for companies to pollute. [?](#)
- Carbon offsetting is an accessible and effective climate action. The more people do it the bigger the impact.